

PROJECT AND TEAM INFORMATION

Project Title

Personalized DSA Learning Path Optimizer Using C and C++ OOP

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Data Structures and Algorithms is an important part of computer science, but many students find it difficult to decide what to study first and what to study next. Everyone does not have the same level of understanding. Some students may be comfortable with Arrays and Linked Lists but struggle with Recursion or Trees, while others may already know the basics and want to move to advanced topics. Our project aims to solve this problem by creating a Personalized DSA Learning Path Optimizer. The system will take information about a student's current knowledge, quiz scores, weak topics, and available study time. Based on this information, it will suggest a suitable sequence of DSA topics.

We will represent DSA topics as nodes of a graph and connect them according to their prerequisites. For example, a student should understand basic concepts before moving to topics such as Trees or Graphs. The system will use graph algorithms to make sure that these dependencies are followed.

The main idea is to make DSA learning more organized and personalized. Instead of giving every student the same fixed learning order, our system will recommend a path according to the student's current performance and requirements.

This project will also allow us to practically apply the DSA and C++ OOP concepts that we are learning in our course.

State of the Art / Current solution

At present, students generally learn DSA through YouTube videos, online courses, books, coding websites, and predefined DSA sheets. Most of these resources already provide a particular order of topics, such as Arrays, Linked Lists, Stacks, Queues, Trees, Graphs, and Dynamic Programming.

Some platforms provide quizzes and track student progress, but students still have to decide which topic they should focus on next. A student who is weak in one particular topic may continue following the same sequence as everyone else.

Our proposed system will focus more on this individual requirement. It will consider the student's performance and prerequisites before suggesting the next topic. For example, if a student has a low score in Linked Lists, the system can give it higher priority before suggesting advanced topics that depend on it.

Therefore, instead of only providing DSA resources, our project will focus on creating a personalized and prerequisite-based learning path.

Project Goals and Milestones

The main goal of our project is to develop a C++ application that can analyze a student's DSA knowledge and generate a suitable learning path.

Goals

- Create a list of important DSA topics.
- Define prerequisites between different topics.
- Represent the topics and their relationships using a graph.
- Take student scores and current knowledge as input.
- Identify weak and strong topics.
- Consider the student's available study time.
- Generate a suitable order for learning the topics.
- Give higher priority to topics that need more attention.
- Store and display the student's learning progress.

Milestones

- Milestone 1: Finalize DSA topics and their prerequisite relationships.
- Milestone 2: Design the graph and decide the data structures required.
- Milestone 3: Create C++ classes for Student, Topic, and Learning Path.
- Milestone 4: Implement the graph and Topological Sorting.
- Milestone 5: Implement topic prioritization using a priority queue.
- Milestone 6: Connect the student's scores and available study time with the recommendation system.
- Milestone 7: Generate and display the personalized learning path.
- Milestone 8: Test the system using different student profiles and improve the results.

Project Approach

We plan to develop the project using C++, mainly using Object-Oriented Programming and Data Structures concepts. First, we will create a collection of DSA topics such as Arrays, Linked Lists, Stacks, Queues, Recursion, Trees, Graphs, Sorting, Searching, and Dynamic Programming. Each topic will contain information such as its difficulty level, estimated study time, and prerequisites.

The prerequisite relationships will be represented using a directed graph. An adjacency list will be used to store the connections between topics.

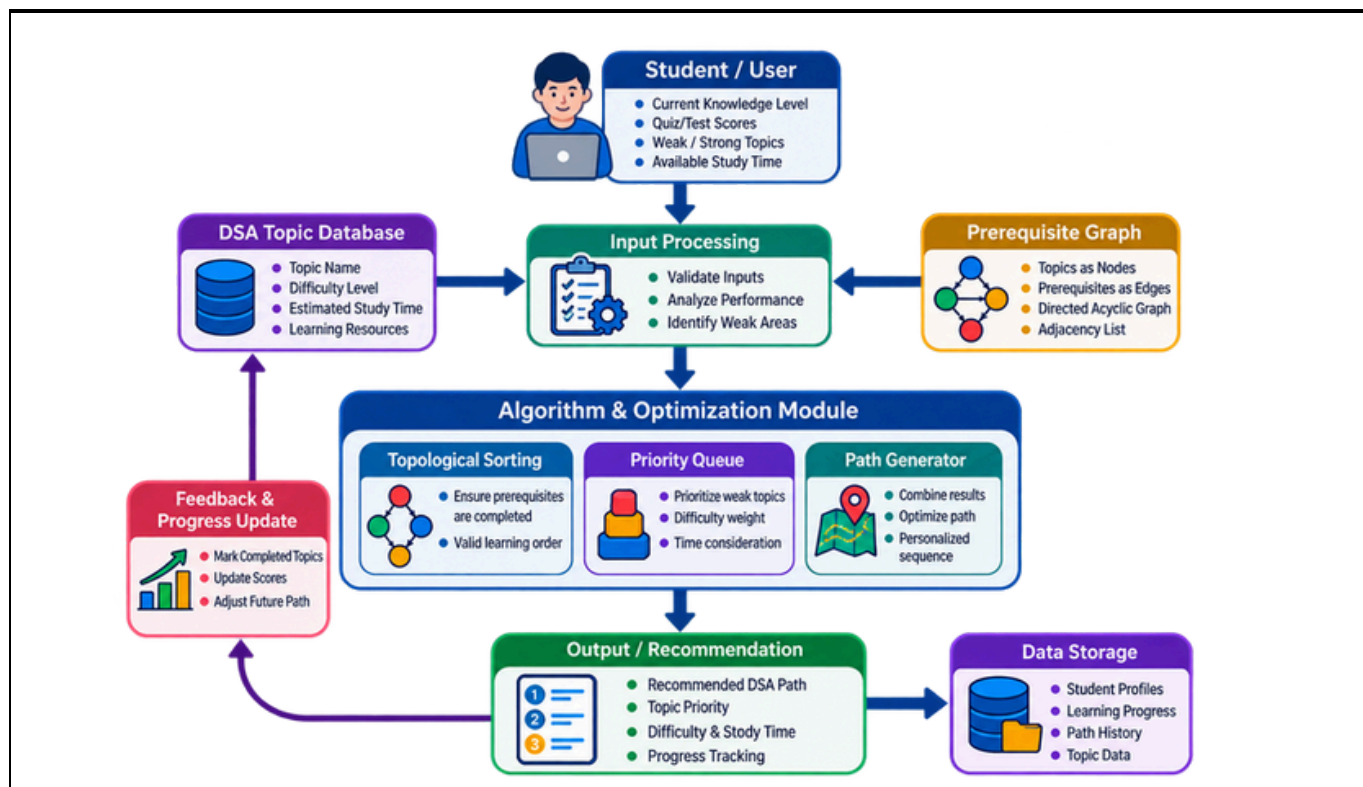
For generating the learning order, we will use Topological Sorting so that prerequisite topics are completed before dependent topics. A priority queue will be used to give more importance to topics where the student has performed poorly or which are important for upcoming topics.

We will create separate C++ classes such as Student, Topic, and LearningPath. This will allow us to demonstrate concepts such as classes, objects, encapsulation, constructors, and member functions.

C++ STL containers such as vector, map, queue, and priority_queue will be used where required. We also plan to use file handling to save student information and progress.

Initially, the project will be developed as a console-based application. If time permits, we will add a simple graphical interface or visualization of the DSA learning graph.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

The major deliverables will include:

- *A DSA topic database.*
- *A prerequisite-based graph of DSA topics.*
- *Student input and performance analysis.*
- *Personalized learning path generation.*
- *Topological Sorting implementation.*
- *Priority-based topic recommendation.*
- *Student progress tracking.*
- *File handling for storing information.*
- *Console-based user interface.*
- *Documentation explaining the algorithms, C++ concepts, system design, and testing.*

For example, if a student performs poorly in Linked Lists and has not yet learned Trees, the system should consider the prerequisite relationship and recommend Linked Lists before moving toward advanced topics.

If sufficient time is available, we will also add a visual representation of the learning graph and the recommended path.

Assumptions

We assume that the DSA topics and their prerequisite relationships are predefined by the project team. Student scores, knowledge levels, and available study time are entered correctly. Topic difficulty and estimated study time are also predefined. The initial version will focus on commonly studied DSA topics. The generated learning path will be a recommendation based on the available input and will not guarantee a particular learning outcome.

References

- *Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein — Introduction to Algorithms.*
- *Mark Allen Weiss — Data Structures and Algorithm Analysis in C++.*
- *GeeksforGeeks — Data Structures, Graph Algorithms, Topological Sorting, Priority Queue and C++ concepts.*
- *Programiz — C++ programming, Object-Oriented Programming and Data Structures.*
- *C++ Reference — C++ Standard Library and STL documentation.*